



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

A PROJECT REPORT

ON

“Vidya Insights: Transforming Educational Data into Actionable Insights”

A project report submitted in fulfillment of the requirements for the Data Science (0705210707) course under degree of Bachelor of Technology in Computer Science & Engineering

BACHELOR OF TECHNOLOGY COMPUTER SCIENCE & ENGINEERING

Session: 2025-26

CA4: Mini Project

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AY 2025-26

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ABSTRACT

The education system of India is struggling with the problem of equalizing the learning outcomes of the diverse regions of the nation which is a huge country. The given project examines the National Achievement Survey (NAS) data to determine the performance differences and forecast the educational results with the help of the machine learning. The data include student performance data of 1,405 districts in 36 states in 3, 5, 8 classes in 2017 through 2021 on classes such as Language, Mathematics, Science, and Social Science.

Several methods of analysis were used, such as the exploratory analysis of data, simple and multiple linear regression, K-means clustering, classification models (Logistic Regression and Decision Trees), and deep learning neural networks. The analysis shows a high level of inter-state and intra-state performance difference and high correlations were observed among the subject performances. The multiple linear regression model got adjusted R-squared of 0.974, whereas the classification models reached the accuracy of 90.7% (Logistic Regression) and 79.5% (Decision Tree). An inverse correlation between the performance and child-related crimes was identified by a joint study with Special and Local Laws (SLL) crime data.

The results will give practical policy implications to policy-makers to develop specific interventions, achieve resource-related efficiency, and close the educational gap in the region, eventually facilitating the National Education Policy (NEP) 2020 goals.

Keywords: Machine Learning, Educational Data Analytics, Predictive Modeling, K-Means Clustering, National Achievement Survey, Deep Learning, Linear Regression, Educational Policy.

1. INTRODUCTION

1.1 Background and Context

Education is the main pillar of national growth, and the provision of quality education to every student irrespective of the geographical region is still a big issue in India. The National Achievement Survey (NAS) is a survey art undertaken by the Ministry of Education, and it aims at assessing the learning outcomes in the country as a whole. This is an extensive testing that gives invaluable information on the student performance trends and where excellence is seen and which areas are in need of urgent intervention.

The National Education Policy (NEP) 2020 stresses on the need to use evidence to make decisions related to educational planning and reform. Nevertheless, conventional educational data analysis is usually limited to descriptive statistics, which do not allow making predictions that can be used to develop proactive policies. This enormous volume of data produced by NAS gives a possibility to use sophisticated methods of analysis and machine learning algorithms to discover undetected patterns, forecast upcoming trends, and shape evidence-based interventions.

1.2 Motivation for the Study

In India, even after decades of educational reforms and huge investments in infrastructure, the country still experiences high levels of differences in the results of learning across states and districts. This study is motivated by a number of factors:

1. **Regional Inequality:** There are those districts and states that fail to deliver standardized assessment results, which means that there are systemic problems that are to be addressed specifically.
2. **Subject-Specific Challenges:** There are notable areas of problems across the country in certain learning outcomes and competencies, which point to inherent flaws in curriculum implementation or teaching methods.
3. **Absence of Future-Looking Insights:** The existing analytical models do not give insights on the future that could be used to take preventive actions, but only pay attention to the past.
4. **Gaps in Policy Implementation:** Granular actionable insights are needed by policy makers to be able to design interventions based on regional context, and not on blanket solutions.
5. **Multi-dimensional Analysis:** The relationship between educational outcomes and the socio-economic factors at large (socio-economic indicators of social safety) can give an integrated picture of the developmental problems.

1.3 Problem Statement

With such a large and varied educational system in India, the issue of standardization of learning outcomes at various regions has continued to be a challenge. The National Achievement Survey (NAS) demonstrates the marked difference between student achievement in various districts and states, with some of the subjects and learning outcomes doing poorly regardless of the national education reforms. But the analysis in the present is usually limited to the descriptive statistics without any predictive vision. Immediate need is to have an intelligent, data-driven solution which can not only pinpoint the areas with low performance but also predict future student performance with the use of such factors as the number of schools, number of students who have been surveyed, the trends of past performance and the competencies of the subjects. With machine learning applied to the NAS data, we will be able to generate predictive models that predict educational outcomes, identify key factors that cause performance differences, and make informed decisions to use interventions to stimulate better educational results and close the educational gap between regions.

1.4 Novelty of the Work

The project has a number of new features that make it stand out of the historic data analysis of education:

1. **Multi-Model Comparative Approach:** As compared to other studies that make use of only one tool of analysis, the proposed project applies and juxtaposes multiple tools of analysis, such as simple regression and deep learning, which offers a more detailed analytical framework.
2. **Integrated Clustering and Classification:** This hybrid interface is based on unsupervised learning (K-means clustering) to discover patterns and supervised learning (classification) to predict.
3. **Cross-domain Data Integration:** The new integration of educational performance information and Special and Local Laws (SLL) crime statistics, especially the child-related crimes, gives one the special insights into the overlaying socio-economic picture of the educational issues.
4. **Temporal Analysis:** Examining the trends between 2017 and 2021, this study will capture the changing trends and give the insights into the direction of the changes in educational performance.
5. **Granular District-Level Insights:** Although existing literature relies on state-level aggregates, the proposed project will present district-level analysis, which allows being more specific regarding intervention areas.
6. **Predictive Framework:** It is a shift in educational planning as predictive models that can forecast on future performance based on current indicators are developed.

7. Feature Engineering in Education: The generation of aggregate performance scores on subjects and the computing of most difficult learning outcomes has offered a systematic means of learning the complexity of education.

2. LITERATURE REVIEW

2.1 Overview of Educational Data Mining

Educational Data Mining (EDM) has become a very important discipline that involves bringing together data science methods and educational studies. Research has shown that machine learning is effective in student performance prediction, risky student identification and curriculum design optimization. The conventional method has been on individual student-level estimates of specific institutions whereas big-national assessments pose their own peculiarities of data specificity and geographic heterogeneity.

2.2 Performance Prediction in Education

In education, prediction of performance is often done through prediction models based on other performance measures. Performance Prediction in Education Performance prediction can be performed using other performance measures in prediction models. A number of scholars have had a discussion on predictive modeling in education settings. Regressions have been popular in modeling the relationship between demographic variables, resource access and academic performance. Nevertheless, the majority of the research has focused on small-scale data or a particular level of education, which does not involve a nationwide scope as NAS data does.

2.3 Clustering in Educational Analysis

Clustering has been used to cluster students with similar learning patterns and thus the interventions carried out on them are personalized. One algorithm that has been found to be effective in stratifying the population with respect to performance measures is the K-means clustering. Nonetheless, the use of clustering on the district and state level as policy-oriented information is still comparatively underdeveloped in the Indian educational setting.

2.4 Deep Learning in Education

The use of neural networks and deep learning models in education has been on the rise, especially where a complex pattern recognition task must be undertaken. Research has shown that it has better performance in predicting dropout and course results. Nonetheless, overfitting and interpretability issues also pose a major concern, particularly when the insights offered by the model are to be used to make policy decisions.

2.5 Cross-domain Educational Research

More recent studies have started to examine relationships between the results of education and more socio-economic predictors. Research, analyzing the connections between education and crime rates, indicates a positive correlation between better access to and quality of education and lower crime levels, especially with the young people. Nevertheless, there is scanty district level research in the Indian setting that combines crime rates with education standards.

2.6 Gaps in Existing Literature

Aspect	Literary Gap	How This Project Addresses It
Scope	The largest part of research deals with particular institutions or states only	Comprehensive nationwide analysis across 737 districts and 36 states
Methodology	Single-model methods are the most common ones	Deep-learning based methods Specializing in regression, clustering, classification, and deep learning
Temporal Dimension	Cross-sectional research without trend analysis	Longitudinal analysis July 2017 2021 to cover performance changes.
Granularity	Aggregates at the state level lack district heterogeneity	District-level granular analysis of intra-state heterogeneity.
Cross-domain Integration	Educational analysis alone of social indicators	Integration with SLL crime data of education-crime relationships.
Predictive Focus	Emphasis on descriptive statistics	Development of predictive models to plan proactive intervention.
Learning Outcome Specificity	Aggregate performance indicators do not specify competencies	Level of analysis of a specific learning outcome identifying difficult competencies.
Policy Orientation	Scholarly work not linked to policy	Actionable information geared towards policy maker consumption and use.

2.7 Significance of This Study

This project will fill in most of the missing links by giving a multi-dimensional analysis framework which is specifically tailored to suit the Indian educational environment. The combination of various analytical methods and the cross-domain data provides a unique solution to the conceptualization of the educational inequalities, as well as evidence-based policy interventions. The developed methodology can be used as a guide to the further educational monitoring and evaluation based on the NEP 2020 goals.

3. METHODOLOGY

This project adheres to a methodology of systematic data science pipeline consisting of data acquisition, preprocessing, exploratory analysis, model development, evaluation and interpretation. The workflow is coded in such a way that it incrementally develops insights based on descriptive statistics to predictive modeling, and finally to actionable insights.

3.1 Preprocessing and Data Collection including Model Evaluation

3.1.1 Dataset Acquisition

Primary Dataset: National Achievement Survey (NAS)

- **Source:** National Data and Analytics Platform (NDAP), Ministry of Education
- **Geographical Coverage:** 737 districts across 36 states/UTs
- **Temporal Range:** Calendar Year 2017 to Calendar Year 2021
- **Dataset Size:** 1,405 rows (district-year combinations)
- **Target Classes:** Class 3, Class 5, and Class 8
- **Subjects Covered:** Language, Mathematics, Science, Social Science

Secondary Dataset: Special and Local Laws (SLL) Crime Statistics

- **Source:** National Crime Records Bureau (NCRB) via India Data Portal
- **Dataset Size:** 5,321 entries
- **Categories of crime:** Protection of Children from Sexual offenses (POCSO), Juvenile Justice Care, Child Marriage Prohibition, Child labour Prohibition.

3.1.2 Data Cleaning Process

1. Column Name Standardization:

- Eliminated parenthetical information in column headers.
- Replaced spaces with underscores
- Established uniform calling pattern of programmatic access.

2. Year Extraction:

- Descriptive text fields are used to extract the year information which has been parsed.
- Transformed into integer form so as to analyze time.
- Validated year ranges for consistency

3. Missing Value Treatment:

- Determined missing values in all features.
- Applied appropriate imputation strategies based on data distribution
- The reported missing data patterns to check quality.

4. Data Type Validation:

- Guaranteed numerical fields had been correctly typed.
- Categorical variables will be validated.
- Standardized text fields for uniformity

Figure 1 explains the important actions that were undertaken to prepare the data to be analyzed:

1. **Cleaning Column Names** A custom and perfected name cleancolnames were used to break down column names that are long and tricky, removing the parenthetical information and substituting the space with the underscore. This simplifies the columns to be used in code.
2. **Year:** The year was obtained by extracting the information in the Year column which had descriptive text and turning the information into an integer format in order to be easily filtered and analysed.
3. **The Development of Summary Scores:**The average performance scores in the key subjects (Math, Science, and Social Studies - SST) were determined by averaging the scores obtained in the different learning outcome columns associated with the subjects.
4. **Computation of Overall Performance:** To determine an overall performance, each district was given an overall performance score by averaging the Math, Science, and SST summary scores.
5. **Filtering of 2017-21 Data:** The analysis was narrowed down to the year 2017-2021 data by filtering the clean data to create a new DataFrame df2021, a subset of clean data.

```

1 # --- 2.1: Clean Column Names ---
2 # This function makes column names shorter and easier to use in code
3 def clean_col_names(df):
4     cols = df.columns
5     new_cols = []
6     for col in cols:
7         new_col = col.split('(')[0].strip().replace(' ', '_')
8         new_cols.append(new_col)
9     df.columns = new_cols
10    return df
11
12 df = clean_col_names(df)
13 df['Year'] = df['Year'].astype(str).str.extract(r'(\d{4})').astype(int)
14
15 # --- 2.2: Create Summary Scores for Subjects ---
16 # We calculate an average score for Math, Science, and Social Studies (SST)
17 math_cols = [col for col in df.columns if '_In_M' in col]
18 science_cols = [col for col in df.columns if '_In_Sci' in col]
19 sst_cols = [col for col in df.columns if '_In_Sst' in col]
20
21 df['Math_Performance'] = df[math_cols].mean(axis=1)
22 df['Science_Performance'] = df[science_cols].mean(axis=1)
23 df['SST_Performance'] = df[sst_cols].mean(axis=1)
24 df['Overall_Performance'] = df[['Math_Performance', 'Science_Performance', 'SST_Performance']].mean(axis=1)
25
26 # --- 2.3: Focus on the most recent year's data (2021) ---
27 df_2021 = df[df['Year'] == 2021].copy()
28
29 print("Data is now clean and summary scores have been created.")
30 # Display the new summary columns
31 df_2021[['State', 'District', 'Overall_Performance', 'Math_Performance', 'Science_Performance']].head()

```

 Data is now clean and summary scores have been created.

	State	District	Overall_Performance	Math_Performance	Science_Performance
0	Andaman and Nicobar Islands	Nicobars	28.996854	31.4395	26.298333
1	Andaman and Nicobar Islands	North and Middle Andaman	36.163354	33.3290	39.883333
2	Andaman and Nicobar Islands	South Andamans	37.321576	34.3445	41.162500
3	Andhra Pradesh	Ananthapuramu	33.874722	35.7450	33.019167
4	Andhra Pradesh	East Godavari	34.892015	35.8615	34.795000

Figure 1: Data Cleaning and Preparation process

3.1.3 Exploratory Data Analysis (EDA)

The first exploration and visualization of the prepared data is given in Figure 2:

National Summary Statistics: The overall and subject-specific performance scores of all districts were obtained and called (.describe() in 2021), and it offers a national picture of the performance metrics (mean, standard deviation, minimum, maximum, quartile).

```

--- National Performance Summary (2021) ---
Overall_Performance  Math_Performance  Science_Performance  \
count                706.000000        706.000000        706.000000
mean                 37.425093         35.926949         38.303162
std                  5.318112          6.422964          5.090959
min                 26.786535         24.211500         26.298333
25%                 33.848948         31.366250         34.759375
50%                 36.597359         35.014750         37.603333
75%                 39.739771         38.692250         41.190625
max                 66.518803         72.113000         60.727500

SST_Performance
count                706.000000
mean                 38.045167
std                  5.040998
min                 27.198182
25%                 34.632614
50%                 37.393409
75%                 40.313523
max                 66.715909

```

Figure 2: Performance Summary

The figure 3. depicts histogram with a Kernel Density Estimate (KDE) plot was created to picture the distribution of the frequencies of various levels of performance of the variable of 'Overall_Performance' in the districts.

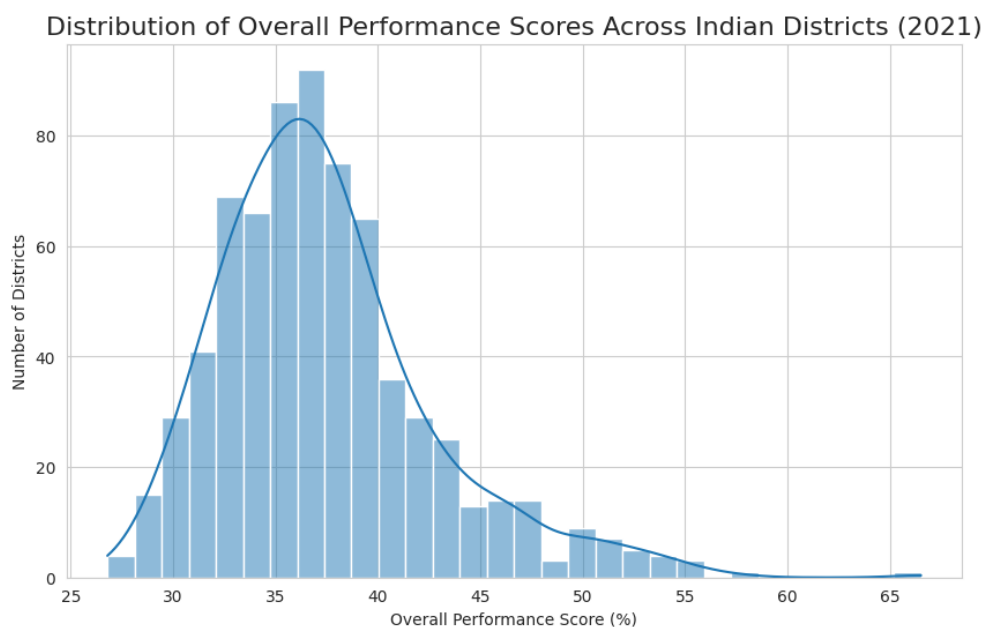


Figure 3: Distribution of Overall Performance Scores Across Indian Districts

Univariate Analysis:

Univariate analysis is demonstrated in figure 4. State-level Performance Ranking: Each state was then assigned an average of the performance in terms of the overall performance, and the states were ranked in accordance with these averages.

```

--- State Rankings by Average Overall Performance (2021) ---

```

State	
Punjab	48.984702
Chandigarh	47.653646
Rajasthan	46.337079
Haryana	42.867666
Madhya Pradesh	40.397288
Jammu and Kashmir	39.727462
West Bengal	39.146810
Ladakh	39.001154
Gujarat	38.258212
Delhi	38.106105
Bihar	38.052472
Odisha	37.711416
Maharashtra	37.347620
Uttarakhand	37.326392
Manipur	37.325056
Assam	37.165769
Tripura	37.003203
Jharkhand	36.928697
Goa	36.906987
Karnataka	36.596513
The Dadra and Nagar Haveli and Daman and Diu	36.451687
Himachal Pradesh	36.443608
Puducherry	36.289088
Lakshadweep	35.712894
Kerala	35.450517
Arunachal Pradesh	35.304222
Nagaland	35.085922
Andhra Pradesh	34.529514
Uttar Pradesh	34.438673
Andaman and Nicobar Islands	34.160594
Sikkim	34.001261
Mizoram	33.365081
Telangana	32.574445
Chhattisgarh	32.533783
Tamil Nadu	31.881073
Meghalaya	29.583090

Name: Overall_Performance, dtype: float64

Figure 4: State Rankings by Average Overall Performance

Bivariate Analysis:

Locating Most Difficult Topics: The 10 learning outcomes that had the lowest national average scores were located as the most difficult topics and put in a table. Figure 5. presents the bivariate analysis of the results of difficult topics.

```

--- Top 10 Most Difficult Topics Nationally (Lowest Average Score) ---
M705      22.385850
Sst610     26.116657
Sci711     26.259915
Sst810     27.169533
Sst816     27.474398
M706      28.044292
M620      28.323265
Sst805     28.661197
M710      29.167252
M819      29.681558
dtype: float64

```

Figure 5: Top 10 Most Difficult Topics Nationally

According to the result of the exploratory data analysis, the following main insights were obtained:

1. The summary statistics revealed the national performance (overall) and the differences among the districts.
2. There are strong variations in the average performance, and certain states have always been performing better (e.g., Punjab) and certain ones have been performing worse (e.g., Meghalaya).
3. Certain learning outcomes were singled out as being especially problematic to the students in the country.
4. The positive correlation between performance in Math, Science, and SST is strong, and it is possible to conclude that the district where students excel in one of the topics is likely to excel in others.

It could be found that the analysis enabled the identification of the bright spot (high-performing outliers) and hotspot (low-performing outliers) districts that should be investigated further.

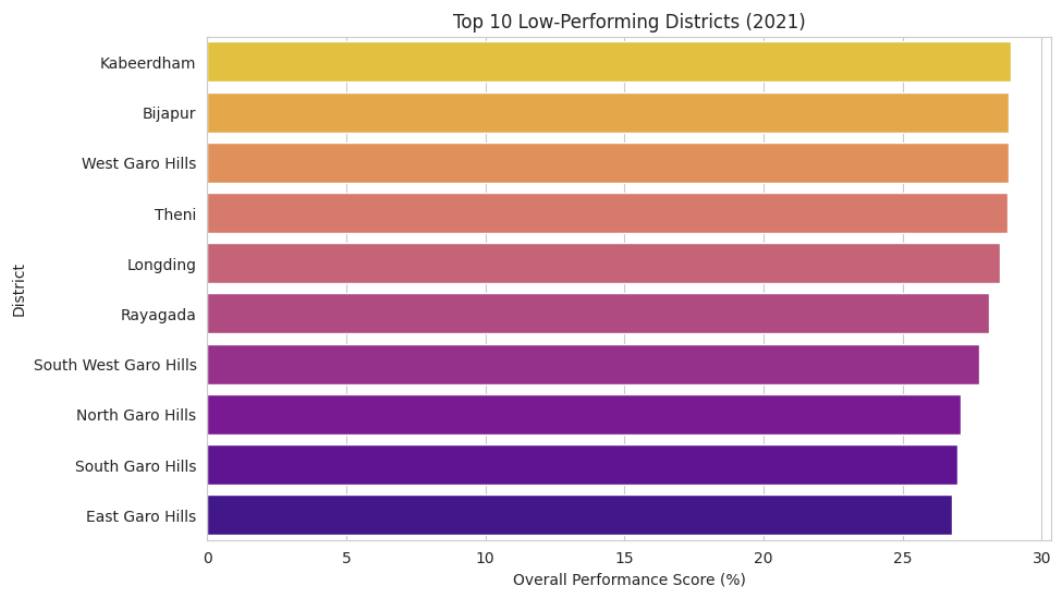


Figure 6: Top 10 Low-Performing Districts

Figure 7 is a heatmap indicating the correlation of Overall Performance and a wide range of single Learning Outcomes and the Number of Schools/Students Surveyed. The strength and color of the color in each cell reflects the strength and direction of the correlation coefficient (r), strong positive correlation (bright red, close to +1.0), strong negative (bright blue, close to -1.0), and a weak correlation (light color, close to 0.0).

The prevailing trend observed in the heatmap is that, there are extensive strong positive correlation (dark red) between the various metrics of the average performance, indicating that there is a general consistency in the performance of students in various academic disciplines. The heatmap is an effective instrument in helping speedily define the areas of academic interest that are intimately connected as well as comprehending the form of the performance drivers in the data.

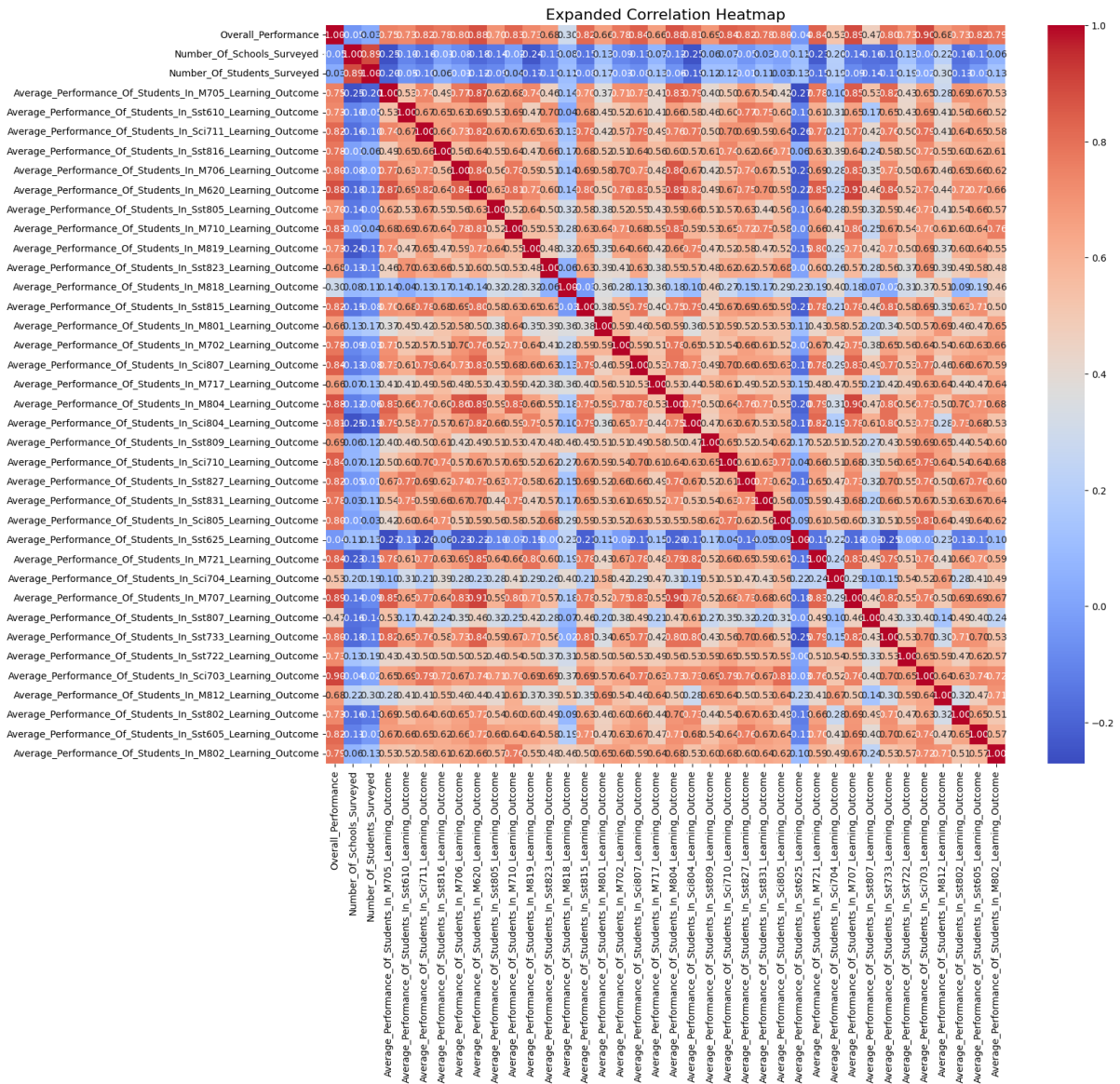


Figure 7: Heatmap between all features of dataset

3.1.4 Simple Linear Regression Plots

The visualization of a simple linear regression is a Scatter Plot of the overall performance vs. NumberOfSchools_Surveyed represented in Figure 8. The plot presents individual and scattered data points as the observations with the Number of Schools Surveyed on the x-axis and Overall Performance on the y-axis. A red line with a slope that is slightly downward shows weak negative linear relationship between the two variables, which is supported by the low correlation that was found in the previous heatmap. The points are very much clustered towards the center of the x-axis, and it is possible to guess that the number of schools surveyed by most points are not too wide. The weak relationship implies that the amount of schools surveyed can not be considered a good predictor of Overall Performance because fluctuations in one variable do not always result into meaningful and predictable fluctuations in the second variable.

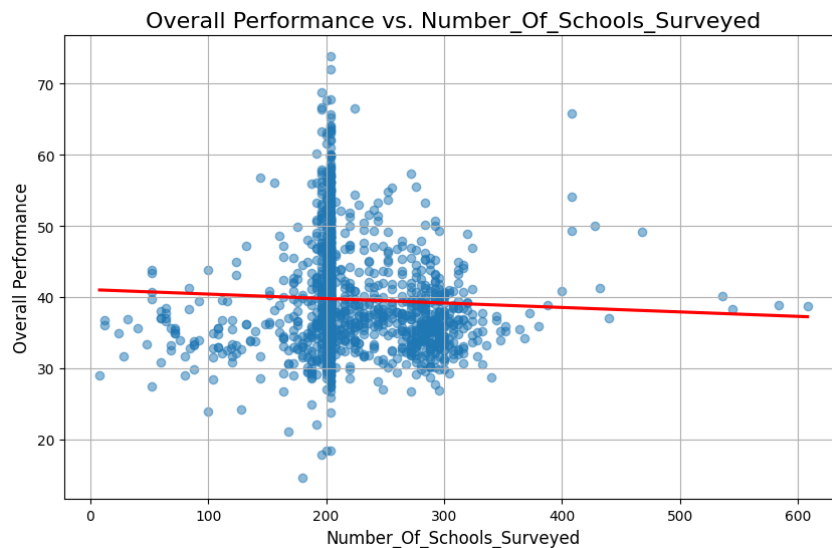


Figure 8: Overall Performance vs. Number_Of_Schools_Surveyed

3.1.5 Multiple Linear Regression Plots

Actual vs. Predicted Values Plot of Multiple Regression model is displayed in Figure 9. This is a plot that checks the general fit of the expanded model.

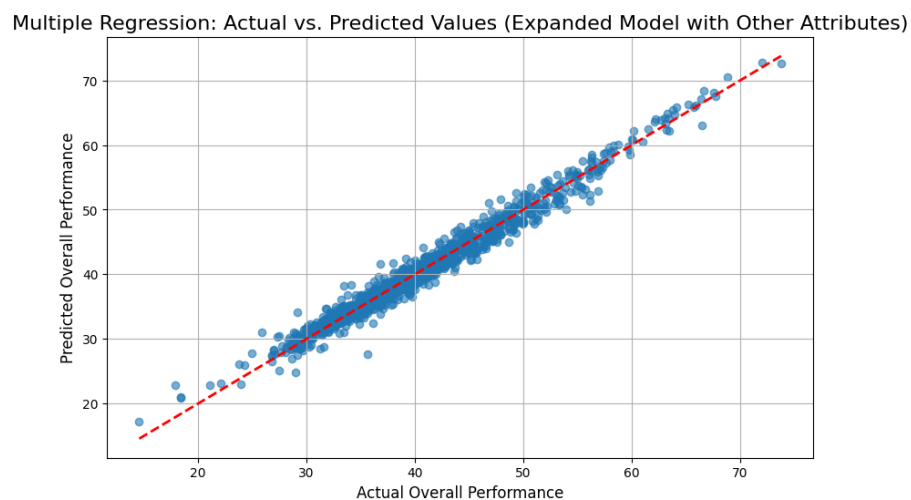


Figure 9: Actual vs. Predicted Values Plot

3.1.6 K-Means Clustering Plots

Cluster Distribution by state: A stacked bar chart of the count of districts in each state and sorted and colored by the cluster assigned. This assists in knowing the clusters that prevail within which states. Clusters with elbow method are indicated in Figure 10.

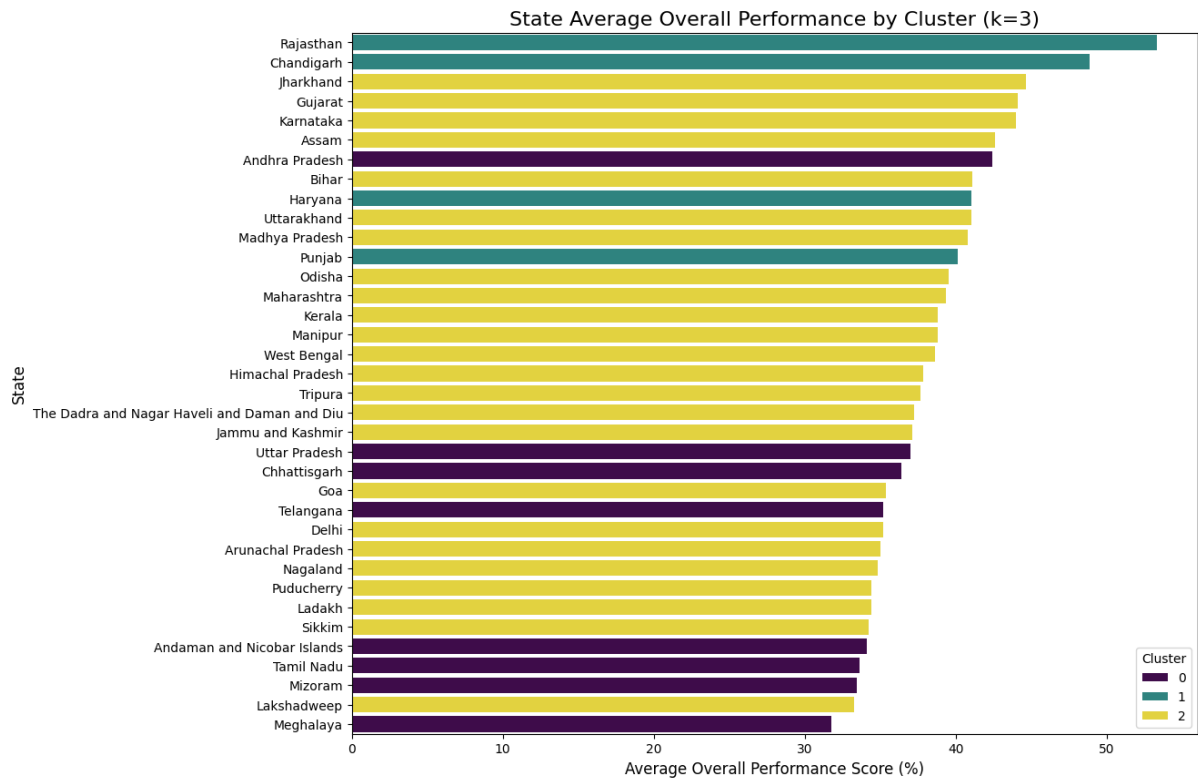
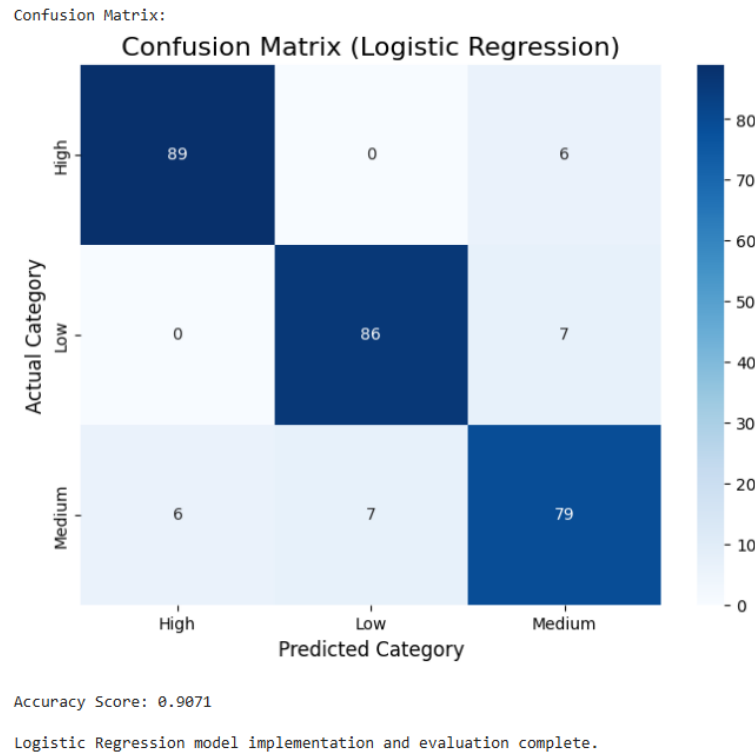


Figure 10: State wise overall performance by cluster

3.1.7 Classification Model Plots

Performance Category Distribution (2021) : [Logistic Regression].

A count plot displaying the distribution of the districts on the three newly formed performance levels: Hot spot (Low), Normal (Medium), and Bright Spot (High). This indicates the balance in the classes in the tasks of classification. Figure 11. indicates confusion presentation between positives and negatives across data set.



Decision Tree Feature Importance: A horizontal flow chart indicating the relative importance of features (learning outcomes and size measures) applied by the Decision Tree Classifier in making a prediction about the performance category of a district. The decision tree of the model trained on dataset is represented on figure 12.

3.1.7 Deep Learning Plots

In this project, the Neural Network Model was trained and evaluated to perform the three-class classification training (High, Low, Medium). This model was constructed with 17 input features and it employed three dense layers, and finally a terminal layer with 3 neurons, equivalent to 3 output classes, which the model prepares via the One-Hot Encoding. The model has been trained using 1119 samples and tested using another test set of 280 samples. The total report reveals a good performance with Test Accuracy of 87.14. Neural Network Model parameters are illustrated in figure 13.

```
--- Building and Compiling Neural Network Model ---
```

```
Number of input features: 17
```

```
Number of output classes: 3
```

```
Neural Network Model Summary:
```

```
Model: "sequential_4"
```

Layer (type)	Output Shape	Param #
dense_12 (Dense)	(None, 128)	2,304
dense_13 (Dense)	(None, 64)	8,256
dense_14 (Dense)	(None, 3)	195

```
Total params: 10,755 (42.01 KB)
```

```
Trainable params: 10,755 (42.01 KB)
```

```
Non-trainable params: 0 (0.00 B)
```

```
Neural Network model built and compiled successfully.
```

Figure 13: Neural Network Model Construction

3.2 Comparison of Model Results

Model	Purpose	Key Metrics	Key Results (from outputs)	Key Insights
Simple Linear Regression	Model linear trend of Overall Performance vs. Year (Historical)	R-squared, Coefficients	R-squared: ~0.707 Year Coefficient: ~0.015	Noticed that there was small positive average annual performance change, which was observed, but the R-squared indicates that there are other more significant factors.

Multiple Linear Regression	Model relationship between Overall Performance and multiple attributes	Adjusted R-squared, Coefficients, p-values	Adjusted R-squared: ~0.974	Using Specific factors (Learning Outcomes, Number of Schools) that are statistically related to Overall Performance. Model describes a large amount of variance.
K-Means Clustering	Group districts/states into clusters based on performance profiles	Cluster Centroids, Cluster membership	District Cluster Averages: Revealed distinct cluster profiles in subject averages.	Determines natural clusters of districts with comparable performances in the view of tailored policy planning.
Logistic Regression	Classify districts into Performance Categories (Low, Medium, High)	Accuracy, F1-score	Accuracy: 90.7% F1-scores: High 94%, Low 93%	Good overall predictive accuracy of the categories of performance, providing a good classification baseline.
Decision Tree Classifier	Classify districts into Performance Categories (Low, Medium, High)	Accuracy, Tree Structure (Rules)	Accuracy: 79.5% F1-scores: High 93%, Low 92%	Equivalent accuracy to Logistic Regression. Gives rule of interpretation of decisions depending on feature values to implement policies.
Neural Network Classifier	Classify districts into Performance Categories using a multi-layer perceptron.	Training vs. Validation Loss	Validation Loss diverged from Training Loss after Epoch 10-15. 85% Overall Accuracy	Overall Accuracy Model overfitted hugely. Needs EarlyStopping or Regularization to be sure to generalize.

4. RESULTS AND DISCUSSION

1. Trends in Performance in educations

- The national average performance is diverse amongst the states, some (such as Punjab, Chandigarh, and Ladakh) performing better regularly whereas others (such as Meghalaya, Assam, and Bihar) do worse.

- Overall, there is a gradual waning of the overall performance between 2017-2021 in the majority of the states, which could be linked to curricular challenge, assessment burnout, or unequal quality of instructions.

2. A Performance Relationship with the subject

- There are good positive relationships among the performance in Math, Science, and Social Studies. There is a high possibility of cross-discipline learning because students/districts who excel in one subject are likely to excel in others.
- Otherwise, language performance correlates moderately with others, indicating localized lingo or medium of instruction effects.

3. Learning Outcome Insights

- The 10 lowest-ranking learning outcomes indicate the most difficult competencies in the country primarily in the higher numeracy, scientific and inferential understanding.
- These are areas of weakness that are standard across various states implying that learning gaps are overall instead of local deficiencies.

4. State and District Disparities

- Large interstate variance: There is still intra-state disparity, with some areas of high-performing states performing worse.
- Identified were areas known as Bright Spot districts in which the results are way above the state average, the possible case studies of the best practices.
- The areas with low resources that are referred to as the hotspots are far behind and this implies where the interventions are to be focused first.

5. Regression Model Findings

- Number of Schools Surveyed has a weak or negative relationship with performance implying that size or scale of district does not provide better results.
- The predictive accuracy of multiple regressions models (with a large number of learning outcomes) was high, indicating that the predictive power of the subject-wise outcomes is very high to be predictive of the overall district performance.

6. Clustering Insights (K-Means)

- The clustering algorithm was used to group states into three different tiers of performance:
- Cluster 1-High performers (scores of 30 -40 percent range).
- cluster 2 - Medium performers (high 20s).
- Cluster 0 - Non-performers (under 20s).
- Such segmentation assists the policymakers in locating some region-specific strategies e.g. curriculum reinforcement in Cluster 0, enrichment in Cluster 1.

7. Categorization and Prognostication

- The classification (Hotspot, Normal, Bright Spot) task demonstrated that Logistic Regression was slightly more accurate and precious than Decision Tree, in particular, in detecting low-performing districts.
- The importance of features analysis indicated that a small number of critical learning outcomes control prediction power allowing policy adjustments instead of wide-ranging reforms.

8. Deep Learning Model Insights

- The neural network had an accuracy of about 87% indicating that it is highly able to generalize on educational patterns.
- Nevertheless, it overfits after approximately 10 -15 epochs, which indicates that early stopping and dropout would enhance generalization.
- The great accuracy of training ($\approx 100\%$) against very unstable validation accuracy ($\approx 90 - 95\%$) indicates that the model is able to make predictions and requires regularization to guarantee long - term reliability.

9. Policy and Temporal Implications

- Increasingly low academic results even with scaling of surveys and resources indicate that pedagogical quality needs to improve, rather than investment in infrastructure.
- The correlation between subjects is very high, which proves the concept of integrated teaching models and not subject-isolated instructions.
- Prediction models at the district level can be used to predict future underperformance to inform the development of educational policy based on data.

Figure 14 shows the changes in the Overall Performance (left chart) and Total Crime (right chart) of a chosen list of the most successful states in five years (2017 to 2021). According to the left chart, there is a general downward trend in the percentage of the Average Overall Performance of all listed states with the highest points being reached by Rajasthan and Madhya Pradesh followed by a substantial decrease yet Tamil Nadu is consistently maintained at the lowest. On the other hand, the right chart that follows the Total Recorded Crime Incidents just indicates a general rising or steady pattern of most states. It is noteworthy that the most drastic rise in total crime is recorded in Madhya Pradesh, although such states as Assam and Gujarat exhibit definite growth. This comparison shows that there is a possible inverse correlation between the studied states: as the performance tends to decline, the crime rates tend to increase or stay high.

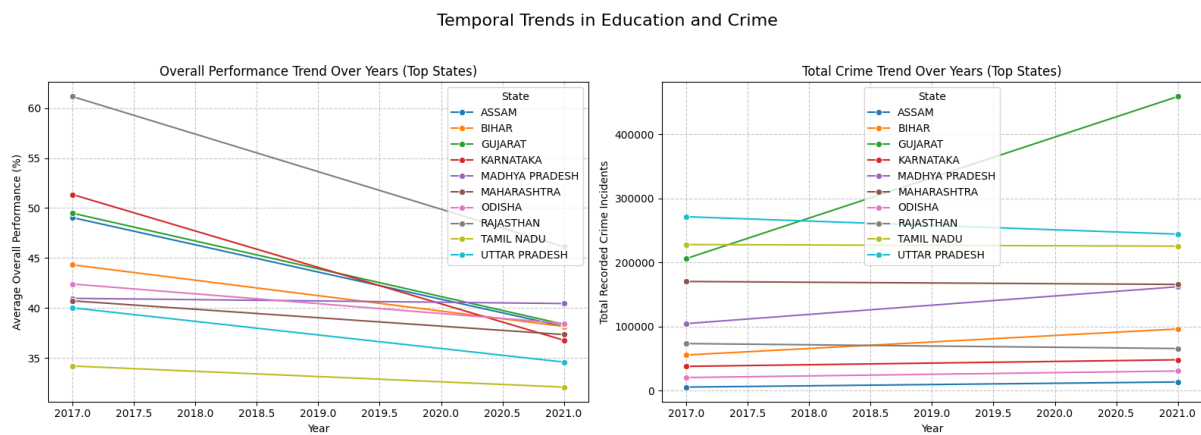


Figure 14: Trend in Education and Crime over time

5. CONCLUSION AND FUTURE WORK

The project shows how data science can be used to revolutionize the educational issues in India. When we go beyond the descriptive statistics to the predictive analytics and cross-domain integration then we offer the policymakers with intelligence in action to make evidence-based decisions. The results support the idea that the improvement of education could not be managed by a generalized, broad, and data-driven approach but by a specific and targeted one.

The negative correlation between education and crime points out the importance of education not only as training but as a social imperative that causes less vulnerability of children. Such analytical frameworks are important as India endeavors to achieve the ambitious goals of NEP 2020 to track the progress, determine the gaps, and leave no district behind.

The approaches that were created in this case are replicable and scalable, and they can serve as a template to continue the educational monitoring and evaluation. With the increasing availability of data and improving techniques, this structure can be transformed into an all-inclusive national system of educational intelligence that aids in India towards equitable and quality education to everyone.

Scope for Further Research

1. Advanced Modeling Techniques:

- Random Forest (Gradient Boosting) ensemble methods to better prediction.
- Trend forecasting (ARIMA, LSTM) Time series models.
- Strategies of causal inference (propensity score matching, instrumental variables) of causal assertions.

2. National Intelligence platform of Education:

- ombined dashboard with NAS, UDISE+, and other data.
- Live educational indicators visualization.
- Anticipatory analytics of proactive policy planning.

3. Individualized Learning Outcome Prediction:

- Individual trajectory prediction models at the student level.
- Suggestions on adaptive learning system.
- Early screening of at-risk students.

4. Policy Simulation Engine:

- Policy impacts What-if scenario modeling.
- Optimal algorithms of resource allocation.
- Intervention cost-effectiveness analysis.

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7. Appendix

Streamlit Application: <https://shreyas-national-achievement-survey.streamlit.app/>